

Abstract Interpretation (3)

CSE552 Program Analysis — Lecture 20

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Limitations of Collecting Semantics

- ▶ *Reachable state* collecting semantics
 - ▶ Collects states that are possible when program execution reaches each program point for some input
- ▶ Precisely captures the meaning of programs in a way that allows us to prove soundness of some analyses, e.g., sign analysis
- ▶ However, for other analyses, it is insufficient because it does not capture all the information about how programs execute

Example

```
y = 0;  
if input() {  
    x = 1;  
} else {  
    x = y + 1;  
}  
// {[x: 1, y: 0]}
```

- ▶ Only tells the state in the last line
- ▶ Does not tell which expression is evaluated
 - ▶ Relevant to available expression analysis
- ▶ Does not provide information about how one state at a program point is related to states at other program points

Trace Semantics

- ▶ Expresses the meaning of a program as the set of traces that can appear when the program runs
- ▶ A *trace* is a finite sequence of pairs of program points and states

$$Trace = (Node \times CState)^*$$

$$A^* = \bigcup_{i \in \mathbb{N}} A^i = A^0 \cup A^1 \cup A^2 \cup \dots$$

Concrete Transfer Functions

- We define the semantics of a single CFG node as a function from a concrete state to a set of concrete states

$$ct_v : CState \rightarrow \mathcal{P}(CState)$$

$v : x = e$

$$ct_v(\rho) = \{\rho[x \mapsto z] \mid z \in \text{ceval}(\rho, e)\}$$

Others

$$ct_v(\rho) = \{\rho\}$$

Notation

- We write a transition between configurations as

$$(v, \rho) \rightarrow (v', \rho') \quad \text{if} \quad v' \in csucc(\rho, v) \text{ and } \rho' \in ct_{v'}(\rho)$$

Trace Semantics of Programs

- ▶ $\langle\!\langle P \rangle\!\rangle \in \mathcal{P}(\text{Trace})$
- ▶ $\langle\!\langle P \rangle\!\rangle$ is the set of finite traces that start at the program entry and in each step proceed according to the CFG
 - ▶ We do not require that the traces reach the program exit
- ▶ $\langle\!\langle P \rangle\!\rangle$ is the least solution to the following constraints:
 - ▶ $I \subseteq \langle\!\langle P \rangle\!\rangle$ where

$$I = \{(entry, [x_1 \mapsto z_1, \dots, x_n \mapsto z_n]) \mid z_1, \dots, z_n \in \mathbb{Z}\}$$

- ▶ $(\pi \cdot s \in \langle\!\langle P \rangle\!\rangle \wedge s \rightarrow s') \Rightarrow \pi \cdot s \cdot s' \in \langle\!\langle P \rangle\!\rangle$

Definition with a Monotone Function

► $tf : \mathcal{P}(\text{Trace}) \rightarrow \mathcal{P}(\text{Trace})$

$$tf(T) = I \cup \text{grow}(T)$$

$$\text{where } \text{grow}(T) = \{\pi \cdot s \cdot s' \mid \pi \cdot s \in T \wedge s \rightarrow s'\}$$

► $\langle\!\langle P \rangle\!\rangle = \text{lfp}(tf)$

Example — Trace Semantics

```
// v0
y = 0; // v1
if input() { // v2
    x = 1; // v3
} else {
    x = y + 1; // v4
}
return; // v5
```

$$\begin{aligned}\langle\!\langle P \rangle\!\rangle = & \{ (v_0, [x \mapsto z_x, y \mapsto z_y]) \mid z_x, z_y \in \mathbb{Z} \} \cup \\ & \{ (v_0, [x \mapsto z_x, y \mapsto z_y]) \cdot (v_1, [x \mapsto z_x, y \mapsto 0]) \mid z_x, z_y \in \mathbb{Z} \} \cup \\ & \{ (v_0, [x \mapsto z_x, y \mapsto z_y]) \cdot (v_1, [x \mapsto z_x, y \mapsto 0]) \cdot \\ & \quad (v_2, [x \mapsto z_x, y \mapsto 0]) \mid z_x, z_y \in \mathbb{Z} \} \cup \\ & \{ (v_0, [x \mapsto z_x, y \mapsto z_y]) \cdot (v_1, [x \mapsto z_x, y \mapsto 0]) \cdot (v_2, [x \mapsto z_x, y \mapsto 0]) \cdot \\ & \quad (v_3, [x \mapsto 1, y \mapsto 0]) \mid z_x, z_y \in \mathbb{Z} \} \cup \\ & \{ (v_0, [x \mapsto z_x, y \mapsto z_y]) \cdot (v_1, [x \mapsto z_x, y \mapsto 0]) \cdot (v_2, [x \mapsto z_x, y \mapsto 0]) \cdot \\ & \quad (v_4, [x \mapsto 1, y \mapsto 0]) \mid z_x, z_y \in \mathbb{Z} \} \cup \\ & \{ (v_0, [x \mapsto z_x, y \mapsto z_y]) \cdot (v_1, [x \mapsto z_x, y \mapsto 0]) \cdot (v_2, [x \mapsto z_x, y \mapsto 0]) \cdot \\ & \quad (v, [x \mapsto 1, y \mapsto 0]) \cdot (v_5, [x \mapsto 1, y \mapsto 0]) \mid z_x, z_y \in \mathbb{Z} \wedge v \in \{v_3, v_4\} \end{aligned}$$

Trace Semantics and Reachable State Semantics

- ▶ Their relation can be expressed as a Galois connection:

$$\alpha_t : \mathcal{P}(\text{Trace}) \rightarrow \mathcal{P}(\text{CState})^n$$

$$\alpha_t(T) = (R_1, \dots, R_n) \text{ where } R_i = \{\rho \mid \exists \pi. \pi \cdot (v_i, \rho) \in T\}$$

$$\gamma_t : \mathcal{P}(\text{CState})^n \rightarrow \mathcal{P}(\text{Trace})$$

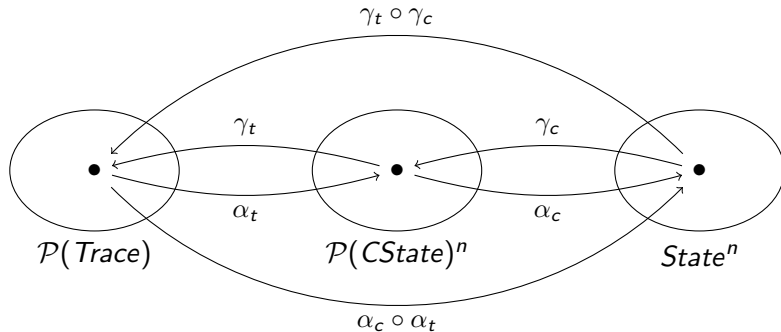
$$\gamma_t((R_1, \dots, R_n)) = \{\pi \cdot (v_i, \rho) \mid \rho \in R_i\}$$

- ▶ Reachable state semantics is an abstraction of trace semantics

Composition of Galois Connections

- ▶ Let $\alpha_1 : L_1 \rightarrow L_2$ and $\gamma_1 : L_2 \rightarrow L_1$ form a Galois connection, and $\alpha_2 : L_2 \rightarrow L_3$ and $\gamma_2 : L_3 \rightarrow L_2$ form a Galois connection
- ▶ Then, $\alpha_2 \circ \alpha_1 : L_1 \rightarrow L_3$ and $\gamma_1 \circ \gamma_2 : L_3 \rightarrow L_1$ form a Galois connection
- ▶ We have a Galois connection between $\mathcal{P}(Trace)$ and $\mathcal{P}(CState)^n$ and a Galois connection between $\mathcal{P}(CState)^n$ and $State^n$
- ▶ We also have a Galois connection between $\mathcal{P}(Trace)$ and $State^n$

Composition of Galois Connections (cont.)



Soundness of Reachable State Semantics

$$\alpha_t(tf(T)) \sqsubseteq cf(\alpha_t(T))$$

- ▶ Let $\alpha_t(T) = (R_1, \dots, R_n)$
- ▶ We need to prove $\alpha_t(tf(T))_i \subseteq cf_i(R_1, \dots, R_n)$
- ▶ Take $\rho \in \alpha_t(tf(T))_i$
 - ▶ Then, $\pi \cdot (v_i, \rho) \in tf(T)$ for some π
- ▶ Case 1: $\pi \cdot (v_i, \rho) \in I$
 - ▶ Then, $v_i = \text{entry}$ and ρ is an initial state
 - ▶ Thus, $\rho \in cf_i(R_1, \dots, R_n)$

Soundness of Reachable State Semantics (cont.)

- ▶ Case 2: $\pi \cdot (v_i, \rho) \in \text{grow}(T)$
 - ▶ Then, $\pi = \pi' \cdot (v_j, \rho')$ where $\pi \in T$ and $(v_j, \rho') \rightarrow (v_i, \rho)$
 - ▶ Since $\pi \in T$, $\rho' \in R_j$
 - ▶ Since $(v_j, \rho') \rightarrow (v_i, \rho)$, $v_i \in \text{csucc}(\rho', v_j)$ and $\rho \in \text{ct}_{v_i}(\rho')$
 - ▶ Thus, $\rho \in \text{cf}_i(R_1, \dots, R_n)$

Soundness of Analysis

► $tf \circ \gamma_t \sqsubseteq \gamma_t \circ cf$

► $cf \circ \gamma_c \sqsubseteq \gamma_c \circ f$

$$\begin{aligned} tf \circ (\gamma_t \circ \gamma_c) &= (tf \circ \gamma_t) \circ \gamma_c \\ &\sqsubseteq (\gamma_t \circ cf) \circ \gamma_c \\ &= \gamma_t \circ (cf \circ \gamma_c) \\ &\sqsubseteq \gamma_t \circ (\gamma_c \circ f) \\ &= (\gamma_t \circ \gamma_c) \circ f \end{aligned}$$

$$lfp(tf) \sqsubseteq (\gamma_1 \circ \gamma_2)(lfp(f))$$

Reaching Definition Analysis

- ▶ $State = (\mathcal{P}(Node), \subseteq)$
- ▶ $JOIN(v) = \bigcup_{u \in pred(v)} \llbracket u \rrbracket$
- ▶ $x = e: \llbracket v \rrbracket = (JOIN(v) \downarrow x) \cup \{v\}$
 - ▶ $\downarrow x$ removes all definitions of x
- ▶ **if** x : $\llbracket v \rrbracket = JOIN(v)$
- ▶ **entry**: $\llbracket v \rrbracket = JOIN(v)$
- ▶ **return**: $\llbracket v \rrbracket = JOIN(v)$

Abstraction Function for Reaching Definition Analysis

- ▶ $def(v, x) = v$ defines x (i.e., v is $x = e$)
- ▶ $last(\pi \cdot s) = s$

$$\alpha_{RD} : \mathcal{P}(\text{Trace}) \rightarrow \mathcal{P}(\text{Node})^n$$

$$\alpha_{RD}(T) = (D_1, \dots, D_n) \text{ where } D_i = \{v_j \mid \pi \cdot (v_j, -) \cdot \pi' \in T \wedge$$
$$def(v_j, x) \wedge$$
$$\forall (v, -) \in \pi'. \neg def(v, x) \wedge$$
$$last(\pi \cdot (v_j, -) \cdot \pi') = (v_i, -)\}$$

Soundness of Reaching Definition Analysis

$$\alpha_{RD}(tf(T)) \sqsubseteq f(\alpha_{RD}(T))$$

- ▶ Let $\alpha_{RD}(T) = (D_1, \dots, D_n)$
- ▶ We need to prove $\alpha_{RD}(tf(T))_i \subseteq f_i(D_1, \dots, D_n)$
- ▶ Take $v_j \in \alpha_{RD}(tf(T))_i$
 - ▶ Then, $\pi \cdot (v_j, \rho_j) \cdot \pi' \in tf(T)$, $def(v_j, x)$, no node in π' defines x , and $last(\pi \cdot (v_j, \rho_j) \cdot \pi') = (v_i, \rho_i)$

Soundness of Reaching Definition Analysis (cont.)

- ▶ Case 1: the trace is in I
 - ▶ Then, $v_i = \text{entry}$, and no such v_j exists
- ▶ Case 2: the trace is in $\text{grow}(T)$
 - ▶ $\pi \cdot (v_j, \rho_j) \cdot \pi'' \in T$ where $\pi'' \cdot s = \pi'$
 - ▶ If $v_j = v_i$, then $\text{def}(v_i, x)$, so $v_j \in f_i(D_1, \dots, D_n)$
 - ▶ Otherwise, $v_j \in D_k$ where v_k is a predecessor of v_i
 - ▶ Since no node after v_j defines x , v_i does not redefine x
 - ▶ Thus, v_j is not removed by $\downarrow x$ and $v_j \in f_i(D_1, \dots, D_n)$

Summary

- ▶ Reachable-state collecting semantics is insufficient for analyses that need information about *how* programs execute, motivating a richer semantics
- ▶ Trace semantics expresses meaning as the set of finite traces (sequences of program-point/state pairs) and is defined as the least fixpoint of $tf(T) = I \cup grow(T)$
- ▶ Reachable-state semantics is an abstraction of trace semantics, connected by the Galois connection (α_t, γ_t)
- ▶ Galois connections compose, yielding a connection from $\mathcal{P}(Trace)$ all the way to the analysis domain $State^n$ and a corresponding soundness argument
- ▶ Reaching definition analysis is naturally justified against trace semantics: its abstraction function α_{RD} and soundness proof rely on tracking which definitions survive along a trace